

Coloring AI - Physiological Personalization MVP

Engineering + Research Specification v0.2

14 August 2026

Executive decision

Build Coloring AI as a privacy-first mobile/desktop PWA that does four things in sequence:

1. **Measure:** use a normal front-facing camera to derive physiology, motion and observed affect locally.
2. **Experiment:** run short randomized repeated visual trials rather than immediately changing the UI based on one emotion classification.
3. **Personalize:** determine which conditions are associated with the user's calmest and most activating response in that session, with explicit confidence and uncertainty.
4. **Export:** turn the winning environment into a portable wallpaper, color recipe and app-specific setup instructions for WhatsApp, Telegram and other tools.

The product's defensible value is not simply camera heart-rate detection. It is the personalized mapping:

USER + CONTEXT + DIGITAL ENVIRONMENT -> MEASURED RESPONSE -> ADAPTIVE VISUAL OUTPUT

What changed in v0.2

This revision incorporates a comparison against an alternative generic webcam-emotion architecture and tightens the production stack.

- React + TypeScript + Vite/PWA is now the recommended application shell.
- MediaPipe Face Landmarker is the default face/behavior layer for a greenfield build.
- MediaPipe Pose Landmarker is explicitly added for torso/shoulder motion and respiration research.
- `face-api.js` / MobileNetSSD is not recommended as the primary new production dependency; the dedicated current MediaPipe Tasks stack and rPPG stack fit the requirements better.
- The product is explicitly **not** "emotion detected -> change CSS." It is "baseline -> randomized repeated stimulus -> quality-controlled multimodal measurement -> within-user ranking -> export."
- A sensor-fusion/consensus layer is now first-class.
- The experiment includes an optional finalist stage instead of brute-forcing all color-font combinations.
- Mobile performance, Web Workers, dependency pinning and privacy auditing are elevated to release requirements.

1. Product thesis

The first impression should be surprising but understandable:

A normal camera can estimate several aspects of your current physiological and behavioral response. Coloring AI then uses a short experiment to learn which visual environment is associated with your calmest response - and gives you that environment to use elsewhere.

The consumer payoff is concrete:

- personalized calming background;
- personalized activating/energizing background;
- optional stress-like result only when multiple signals agree;
- typography recommendation;
- incoming/outgoing bubble colors chosen for readability;

- downloadable wallpaper;
- exact HEX/OKLCH palette;
- WhatsApp/Telegram setup instructions;
- optional Emoji Mirror result.

2. What the product should and should not claim

Good claims

- “camera-derived heart rate estimate”;
- “breathing-rate estimate”;
- “pulse variability / PRV”;
- “observed facial-expression change”;
- “calmest response in this session”;
- “most activating response”;
- “strongest spontaneous mimicry response”;
- “results were consistent across repeated trials”;
- “signal quality was insufficient” when necessary.

Claims to avoid until validated

- medical-grade measurement in uncontrolled conditions;
- ECG-equivalent HRV;
- “deepest breathing” from face rPPG alone;
- direct internal emotion detection;
- stress inferred from HR alone;
- universal statements such as “blue is the best calming color”;
- causal health claims from a single personalization session.

3. Consumer journey

Step	Screen	Purpose
1	Landing	Benefit, privacy, explanation and start CTA.
2	How it works	Explain local camera sensing and randomized personalization.
3	Consent	Explicit camera/local-analysis consent; no medical framing.
4	Camera setup	Center face, stable light, minimal movement, signal-quality gate.
5	Live wow moment	Show HR, breathing estimate, quality and limited motion/expression information.
6	Neutral baseline	Establish within-user reference.
7	Color trials	Five matched-lightness color conditions, repeated in randomized/counterbalanced order.
8	Font trials	Five typography conditions on a neutral color environment.
9	Emoji Mirror	Optional separate mimicry stage.
10	Finalists	Optional top 2 colors x top 2 fonts, not all 25 combinations.
11	Results	Calm, activation, uncertainty/confidence and observed deltas.

Step	Screen	Purpose
12	Export	Wallpaper PNG, palette, bubble colors, typography and app instructions.

4. Recommended production architecture

```

FRONT CAMERA - local only
|
+--> Vital Camera / FacePhys
|   +--> BVP
|   +--> heart rate
|   +--> beat intervals
|   +--> PRV side outputs
|   +--> rPPG SQI
|
+--> MediaPipe Face Landmarker
|   +--> landmarks
|   +--> blendshapes / expression features
|   +--> face transform / pose features
|   +--> blink / mouth / gaze-derived behavior
|
+--> MediaPipe Pose Landmarker
    +--> shoulders / chest / torso motion
    +--> motion-based respiration channel

```

```

SECONDARY VALIDATION
+--> POS / CHROM / PRISM-style estimator

```

ALL SIGNALS

```

|
v
SENSOR HUB / COMMON CLOCK

```

```

|
+--> quality gate
+--> disagreement tracking
+--> local confidence
|
v

```

EXPERIMENT ENGINE

```

+--> seeded randomization
+--> baseline
+--> washout
+--> transition exclusion
+--> repeated trials
+--> finalist confirmation
|
v

```

WITHIN-USER SCORING

```

|
v

```

RESULTS + WALLPAPER + COLOR RECIPE

5. Application stack

Recommended:

- React;
- TypeScript;
- Vite;
- PWA manifest/service worker;
- `getUserMedia()` for camera;
- Web Workers for expensive synchronous vision/signal work;
- CSS custom properties for all experimental visual variables;
- local memory / IndexedDB / localStorage for consumer-session state;
- no backend requirement for the first local-only personalization experience.

Production rules:

- pin reviewed model/runtime versions;
- self-host model/WASM assets where allowed;
- do not pull production ML dependencies from moving `main` branches;
- enforce CSP;
- audit analytics/error tools so they never receive video, BVP or biometric values by accident.

6. Technology assessment

Candidate	Role	Integration	Assessment
Vital Camera SDK + FacePhys	Primary browser physiology	Low-Medium	Best current fit for local HR/BVP/beat/PRV-side outputs plus affect/behavior signals. Public repo is young, so pin/vendor and benchmark.
MediaPipe Face Landmarker	Face/behavior	Low	Current Google web task for landmarks and facial-expression/blendshape output. Preferred over building new product around legacy face-api.js.
MediaPipe Pose Landmarker	Torso/motion	Low-Medium	Current Google web task for body landmarks; appropriate for shoulder/chest motion and respiration channel.
open-rppg / FacePhys	Research harness	Low	Fast Python comparison/validation environment and access to multiple models.
POS / CHROM	Secondary physics baselines	Low	Useful independent estimators and artifact/disagreement signals.
PRISM	Secondary estimator	Medium	Training-free adaptive rPPG candidate with a different methodological path.
CanonicalPhys	Pose hardening	High	Specifically targets rPPG degradation under head pose. Important R&D candidate.

Candidate	Role	Integration	Assessment
PhysMamba / RhythmMamba	Benchmark	High	Strong research comparisons, not the fastest first browser deployment.
EmotiEffLib	Optional affect benchmark	Medium	Lightweight ONNX/Python/C++ expression analysis. Use as observed affect, not internal-emotion truth.
Reperio-rPPG	2026 benchmark watch	High	Public code and strong research claims under motion/illumination; benchmark if reproducible in our harness.

7. Why FacePhys/Vital Camera is the first production candidate

Vital Camera’s public repository currently exposes browser-based heart rate, HRV-related outputs, emotion, gaze, eye state, mouth/speaking and head pose. FacePhys is designed for low-latency edge operation.

Published FacePhys evidence includes:

Evidence	Published result	Interpretation
CPU memory footprint	3.6 MB	Strong edge-deployment profile.
Per-frame latency	9.46 ms	Below a 30 fps frame interval in the reported CPU benchmark.
MMPD intra-dataset HR MAE	5.58 BPM	More realistic/challenging than saturated easy datasets; still not equal to our real-user environment.
RLAP -> MMPD cross-dataset MAE	5.30 BPM	Relevant generalization result.
Cross-dataset VitalVideo MAE	0.77 BPM	Strong reported cross-dataset result under that protocol.
PURE cross-dataset MAE	0.24 BPM	Near-saturated/easier dataset; do not use as sole marketing number.
UBFC cross-dataset MAE	0.43 BPM	Also not sufficient to predict real-phone performance.

These numbers are paper results, not guarantees for our production build. We must benchmark the exact browser implementation, camera pipeline and device matrix against synchronized contact PPG.

8. Accuracy policy

Public accuracy claims require our own validation.

Measure at minimum:

- HR MAE;
- RMSE;
- correlation;
- failure/rejection rate;

- valid-time coverage;
- motion/pose failure modes;
- dynamic-lighting failure modes;
- device/browser subgroup;
- skin-tone subgroup;
- latency, CPU, memory and thermal behavior.

A product that returns a plausible value during bad signal is worse than one that says “measurement unavailable.” Quality rejection is a feature.

9. Face, motion and observed affect

The browser should use MediaPipe Face Landmarker as the default face layer. It can provide face landmarks and face blendshapes/expression-related outputs in web video workflows.

Use the face layer for:

- consistent facial ROI geometry;
- head pose / transformation changes;
- smile-related features;
- blink/eye openness;
- mouth/jaw and talking indicators;
- gaze-like/attention features when supported or derived carefully;
- per-session neutral calibration.

Do not reduce this to “happy/sad/angry = truth.” Continuous and calibrated changes are more useful for our experiment.

10. Respiration and motion

Respiration should eventually use two independent channels:

1. low-frequency respiratory modulation of BVP/rPPG;
2. upper-torso/shoulder motion from MediaPipe Pose Landmarker or optical-flow/ROI processing.

Fusion rules:

- output each channel’s confidence;
- fuse only when they agree within tolerance or one is clearly high-confidence and the other is unavailable;
- reject talking and large motion;
- label amplitude as a motion proxy, not absolute breath depth.

11. Experiment design

Core design: within-subject randomized repeated blocks. The user is their own control.

Quick mode

- neutral baseline: 25 s;
- colors: 5 conditions x 2 blocks x 15 s, 5 s washout;
- fonts: 5 conditions x 2 blocks x 12 s, 5 s washout;
- emoji: 5 conditions x 2 blocks x 10 s, 5 s washout, optional;
- discard first about 4 s after every active visual transition;
- pause/reject low-quality windows.

Optional finalist confirmation

Instead of testing all 25 color-font combinations:

- choose top 2 colors;

- choose top 2 fonts;
- test 4 combinations;
- repeat each combination;
- select a final theme only if repeatability/confidence is adequate.

This keeps a consumer session practical while preserving repeated measurement.

12. Stimulus design

Color

Use approximately matched perceptual lightness in OKLCH. First validation set should avoid extreme saturation.

Font

Use broad typographic categories, with neutral color environment. Normalize size, line height and geometry as much as possible.

Emoji Mirror

Keep background/font/chat content fixed. Do not tell users to imitate. Measure spontaneous expression and physiological response.

Chat content

Neutral, low-arousal, semantically comparable content only. The content should occupy attention without creating the primary emotional effect.

13. The screen-color / rPPG confound

This remains the single biggest scientific risk.

A phone/laptop display changes the light spectrum on the face. rPPG detects subtle reflected-light changes. Therefore a color switch can directly change the camera signal even if heart rate does not change.

Required mitigation:

- perceptually matched-lightness stimuli;
- constant neutral camera/sensor region when possible;
- exact color and luminance logging;
- face/frame RGB tracking;
- transition exclusion;
- neutral washouts;
- SQI and pose rejection;
- secondary estimator;
- contact-PPG validation while screen colors change.

Go/no-go:

If display color alone produces camera HR/BVP shifts comparable to the claimed calming effect while the contact sensor is stable, physiology-based color claims remain disabled.

14. Scoring

Starting transparent heuristic:

$$\text{calm} = 0.45 * (-z(\text{HR})) + 0.30 * (-z(\text{respiration})) + 0.15 * z(\text{PRV}) + 0.10 * z(\text{expression tendency})$$

This is not a validated biomarker.

Rules:

- score only valid post-transition samples;
- compare within user/session;
- down-weight or disable PRV when beat count is inadequate;
- report individual components;
- require repeated-block consistency;
- force uncertainty rather than invent a winner.

Activation can reflect higher HR/respiration with neutral-positive affect. A stress-like output requires increased HR/respiration, reduced PRV and non-positive affect/behavior agreement. HR alone is never sufficient.

15. Confidence

Confidence should combine:

- valid data coverage;
- rPPG SQI;
- valid beat count;
- repeat consistency;
- winner/runner-up margin;
- cross-signal agreement;
- estimator agreement where available;
- motion/illumination warnings.

Suggested result labels:

- high confidence;
- moderate confidence;
- low confidence / exploratory;
- uncertain - repeat session.

16. Privacy and data flow

Default consumer mode:

- raw camera video: local only;
- raw frames: never uploaded;
- BVP: memory only by default;
- physiology/expression inference: local;
- result summaries: local by default;
- wallpaper rendering: local;
- analytics: no biometric values without explicit separate opt-in.

If research sharing is later enabled, use a separate informed-consent flow and upload derived trial summaries rather than video by default.

17. WhatsApp, Telegram and other app portability

WhatsApp

Official Help Center materials confirm chat theme, color and wallpaper customization. The guaranteed export should therefore be:

- wallpaper PNG;
- background color;
- incoming/outgoing bubble color suggestions;
- contrast-safe text color;
- exact color recipe;
- app setup instructions.

Do not promise arbitrary font-family installation into ordinary WhatsApp. Font controls vary by app/account/platform; treat them as conditional.

Telegram

Telegram officially supports custom themes, wallpapers and accent colors. Provide theme/palette instructions and a portable theme artifact only when the current platform specification is confirmed.

Generic exports

- PNG wallpaper;
- palette JSON;
- CSS custom properties;
- design-token JSON;
- copyable HEX/OKLCH values.

18. Result experience

Example structure:

Your calmest response - Quiet Sage

- HR: -4.1 BPM vs session baseline.
- Breathing: -1.8 breaths/min.
- PRV: +9% when signal was valid.
- Expression: neutral-positive maintained.
- Repeated-block consistency: high.
- Confidence: 82%.

Then show:

- wallpaper preview;
- bubble colors;
- typography recommendation;
- “Use this in WhatsApp”;
- “Use this in Telegram”;
- download PNG;
- copy color recipe.

Avoid claiming a universal or permanent physiological truth from one session.

19. Validation plan before public claims

Stage A - Engineering reliability

Browser/device support, permissions, model load, dropped frames, battery/thermal behavior, orientation and back-grounding.

Stage B - Sensor validity

Synchronized webcam vs contact PPG across devices, skin tones, lighting and mild movement. Report error plus rejection/failure rate.

Stage C - Color-confound validity

Change display colors while contact PPG verifies the physiological signal. Quantify device-specific camera artifacts.

Stage D - Repeatability

Repeat the same user on separate days. Measure color/font rank stability and confidence calibration.

Stage E - Actual personalization effect

Only after A-D, compare the individualized winning theme against a neutral/control theme in a new session.

20. Engineering roadmap

Sprint	Deliverable
0	Pin/vendor the sensor stack; reproduce FacePhys/Vital Camera; run first contact-PPG comparison.
1	React/TypeScript/PWA shell; consent; camera; quality gate; live wow metrics.
2	MediaPipe Face + Pose; common SensorHub; Web Workers.
3	Randomized experiment engine; baseline; washouts; transition exclusion; repeats.
4	Results; confidence; wallpaper/palette export; WhatsApp/Telegram instructions.
5	Dual-channel respiration and motion rejection.
6	Screen-light correction and second rPPG estimator; contact-reference study.
7	CanonicalPhys / newer benchmark candidates; pose and illumination hardening.
8	Optional research opt-in and population-level hierarchical analysis.

21. MVP acceptance criteria

Do not call the MVP production-ready until:

- camera permission flow works on target mobile/desktop browsers;
- no raw camera data leaves device;
- model/runtime versions are pinned;
- signal-quality failures are explicit;
- randomization is reproducible and logged;
- trial transitions are excluded correctly;
- results can be uncertain;
- wallpaper/color export works;
- accessibility contrast is enforced;
- synchronized contact-PPG benchmark exists;
- screen-color confound test has a documented result.

22. Primary repositories and sources

- Vital Camera SDK: <https://github.com/KegangWangCCNU/vitalcamera-sdk>
- FacePhys browser demo: <https://github.com/KegangWangCCNU/FacePhys-Demo>
- FacePhys paper: <https://arxiv.org/abs/2512.06275>
- open-rppg: <https://github.com/KegangWangCCNU/open-rppg>
- MediaPipe Face Landmarker for Web: https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker/web_js
- MediaPipe Pose Landmarker for Web: https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker/web_js
- CanonicalPhys code: <https://github.com/infraface/CanonicalPhys>
- CanonicalPhys paper: <https://arxiv.org/abs/2607.15995>
- EmotiEffLib: <https://github.com/sb-ai-lab/EmotiEffLib>
- Reperio-rPPG: <https://github.com/deconasser/Reperio-rPPG>
- WhatsApp themes/colors/wallpaper: <https://faq.whatsapp.com/6298875896807790>
- Telegram themes: <https://core.telegram.org/themes>

Bottom line

The strongest merged architecture keeps the simple strengths of a generic webcam web app - React, browser camera access, local processing and CSS-driven visual changes - but replaces the generic “emotion classifier controls the theme” idea with a substantially stronger product:

dedicated rPPG + current face/pose tracking + quality control + randomized repeated experimentation + within-user multimodal scoring + uncertainty + portable personalized output.

The two scientific gates remain unchanged: validate the exact camera physiology stack against a contact reference, and prove that screen-color changes are not creating false rPPG effects.